

UNIVERSITY OF LISBON

ISEG- LISBON SCHOOL OF ECONOMICS AND MANAGEMENT

REGULAR SEASON EXAM - DECEMBER 2024

ADVANCED TOPICS IN ECONOMETRICS

Module Convenor:
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Duration: THREE HOURS

Use answer booklets of ISEG (folhas de teste do ISEG)

Instructions (please read before starting): Write in a clear legible manner in ink/ballpoint. Do not use pencils or erasable pens. Approved calculators are permitted. Only one sheet (2 sides A4) of notes made exclusively by the student may be consulted (no material distributed by the teacher in any form is allowed). Whenever conducting a test use a 5% significance level unless stated otherwise. Also be sure to state null and alternative hypotheses, null distribution (with degrees of freedom), rejection criterion (critical values and rejection region) and outcome. If you are asked to derive something, give all intermediate steps also. Do not answer questions with a “yes” or “no” only, but carefully justify your answer.

Section A - Topics in Microeconometrics

Question 1

Suppose we use an index formulation for a discrete choice model, but it is felt that the latent variable is strictly positive. This is accommodated by supposing that the latent variable Y^* has exponential density with parameter γ , so the density function of Y^* conditional on the vectors of exogenous variables X and Z is

$$f(y^*|X, Z) = \begin{cases} \gamma^{-1} \exp(-y^*/\gamma) & \text{if } y^* > 0 \\ 0 & \text{if } y^* \leq 0 \end{cases},$$

with $\gamma = \exp(X'\beta)$ and the conditional distribution function is

$$F(y^*|X, Z) = \begin{cases} 1 - \exp(-y^*/\gamma) & \text{if } y^* > 0 \\ 0 & \text{if } y^* \leq 0 \end{cases}.$$

We observe $Y = 1$ if $Y^* > Z'\alpha$ and $Y = 0$ if $Y^* \leq Z'\alpha$. We would like to estimate β and γ . Assume additionally that a random sample is available $\{(Y_i, X_i', Z_i')'\}_{i=1}^n$.

- (a) (1.4 Marks) Give the log-likelihood function for the observed data.
- (b) (1.4 Marks) Let X_j be the element j of X , what is the effect of a one-unit change in X_j on $\mathcal{P}[Y = 1|X, Z]$?
- (c) (1.4 Marks) Suppose that $Y = 1$ if $Y^* > \exp(Z'\alpha)$ and $Y = 0$ if $Y^* \leq \exp(Z'\alpha)$ and $X = Z$. Is there a problem identifying α and/or β ? Explain your answer.

Question 2

Consider the latent variable regression model

$$Y_i^* = \mathbf{X}_i' \boldsymbol{\beta}_0 + u_i, \quad u_i \sim \mathcal{N}(0, \sigma^2),$$

where $\boldsymbol{\beta}_0$ is a k -vector of parameters and $\mathbf{X}_i$ is a k -vector of exogenous regressors. Although Y_i^* is not observable, the following variable is observed:

$$Y_i = \begin{cases} 0 & \text{if } Y_i^* \leq 0 \\ Y_i^* & \text{if } Y_i^* > 0 \end{cases}.$$

In this context, note the following:

$$\begin{aligned} E(Y_i | \mathbf{X}_i, Y_i > 0) &= \mathbf{X}_i' \boldsymbol{\beta}_0 + \sigma \frac{\phi(\mathbf{X}_i' \boldsymbol{\beta}_0 / \sigma)}{\Phi(\mathbf{X}_i' \boldsymbol{\beta}_0 / \sigma)}, \\ E(Y_i | \mathbf{X}_i) &= \mathbf{X}_i' \boldsymbol{\beta}_0 \Phi\left(\frac{\mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right) + \sigma \phi\left(\frac{\mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right), \end{aligned}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the cumulative distribution function and the probability density function of the standard normal distribution respectively.

- (a) (1.4 Marks) Explain why the Ordinary Least Squares regression of Y_i on $\mathbf{X}_i$ using either all observations or just those with $Y_i > 0$ would produce biased and inconsistent estimators.
- (b) (1.4 Marks) Assume that a random sample $\{(Y_i, \mathbf{X}_i')\}_{i=1}^n$ is available. Write down the expression of the log-likelihood function of this model.
- (c) (1.4 Marks) Define a binary variable: $D_i = 1$ if $Y_i > 0$, $D_i = 0$ if $Y_i = 0$. What parameters would be estimated by a Probit model with dependent variable D_i and explanatory variables $\mathbf{X}_i$?
- (e) (1.4 Marks) Show how the Ordinary Least Squares estimator can be “corrected” using a 2-step estimator.

Question 3

Assume that the random variables x and ε are statistically independent, and ε is a continuous random variable with $E[\varepsilon] = 0$. Let q_α denote the α -th quantile of ε . Assume the following quantiles are given: $q_{0.5} = 0.5$, $q_{0.75} = 1.2$, and $q_{0.25} = -0.7$. Construct two new random variables:

$$y = 2 + 0.5x + \varepsilon, \quad z = 2 + 0.5x + x\varepsilon.$$

Define $Q_y(\alpha|x)$ as the α -th quantile of y conditional on x , and $Q_z(\alpha|x)$ as the α -th quantile of z conditional on x .

- (a) (1 Mark) Compute $E(y|x)$, $Q_y(0.5|x)$, $Q_y(0.75|x)$, and $Q_y(0.25|x)$.
- (b) (1.6 Marks) Compute $E(z|x)$, $Q_z(0.5|x)$, $Q_z(0.75|x)$, and $Q_z(0.25|x)$ assuming $x > 0$.
- (c) (1.6 Marks) Compute $E(z|x)$, $Q_z(0.5|x)$, $Q_z(0.75|x)$, and $Q_z(0.25|x)$ assuming x can take any value in $\mathbb{R}$.

Question 4

(1.8 Marks) Consider the model

$$y = x'\beta + u,$$

where x includes unity, and u is independent of x . Now suppose that u follows a logistic distribution (which is symmetric about zero). The density is:

$$f_u(u) = \frac{\exp(-u)}{(1 + \exp(-u))^2},$$

where $E(u) = 0$, $\text{Var}(u) = \pi^2/3$. In this context, note that $E(y|x) = \text{Med}(y|x) = x'\beta$. Which estimator is more efficient, the Ordinary Least Squares or the Least Absolute Deviation estimator?

Section B - Topics in Time Series

Question 5

Suppose that Z_t and W_t follow a VAR(1) process:

$$\begin{bmatrix} Z_t \\ W_t \end{bmatrix} = \begin{bmatrix} 0.7 & 0.2 \\ 0.4 & 0.5 \end{bmatrix} \begin{bmatrix} Z_{t-1} \\ W_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix},$$

where $(\varepsilon_{1t}, \varepsilon_{2t})'$ is a vector white noise process such that:

$$E(\varepsilon_{it}) = 0, \quad \text{var}(\varepsilon_{it}) = \sigma_i^2, \quad i = 1, 2, \quad \text{and} \quad E(\varepsilon_{1t}\varepsilon_{2t}) = 0.$$

- (a) *(1.4 Marks)* Is the process $(Z_t, W_t)'$ stationary?
- (b) Show that W_t follows an ARMA(2,1) process by proving the following:
- (i) *(1.4 Marks)* $W_t = 1.2W_{t-1} - 0.27W_{t-2} + 0.4\varepsilon_{1t-1} + \varepsilon_{2t} - 0.7\varepsilon_{2t-1}$.
- (ii) *(1.4 Marks)* $v_t = 0.4\varepsilon_{1t-1} + \varepsilon_{2t} - 0.7\varepsilon_{2t-1}$ follows a MA(1) process.

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